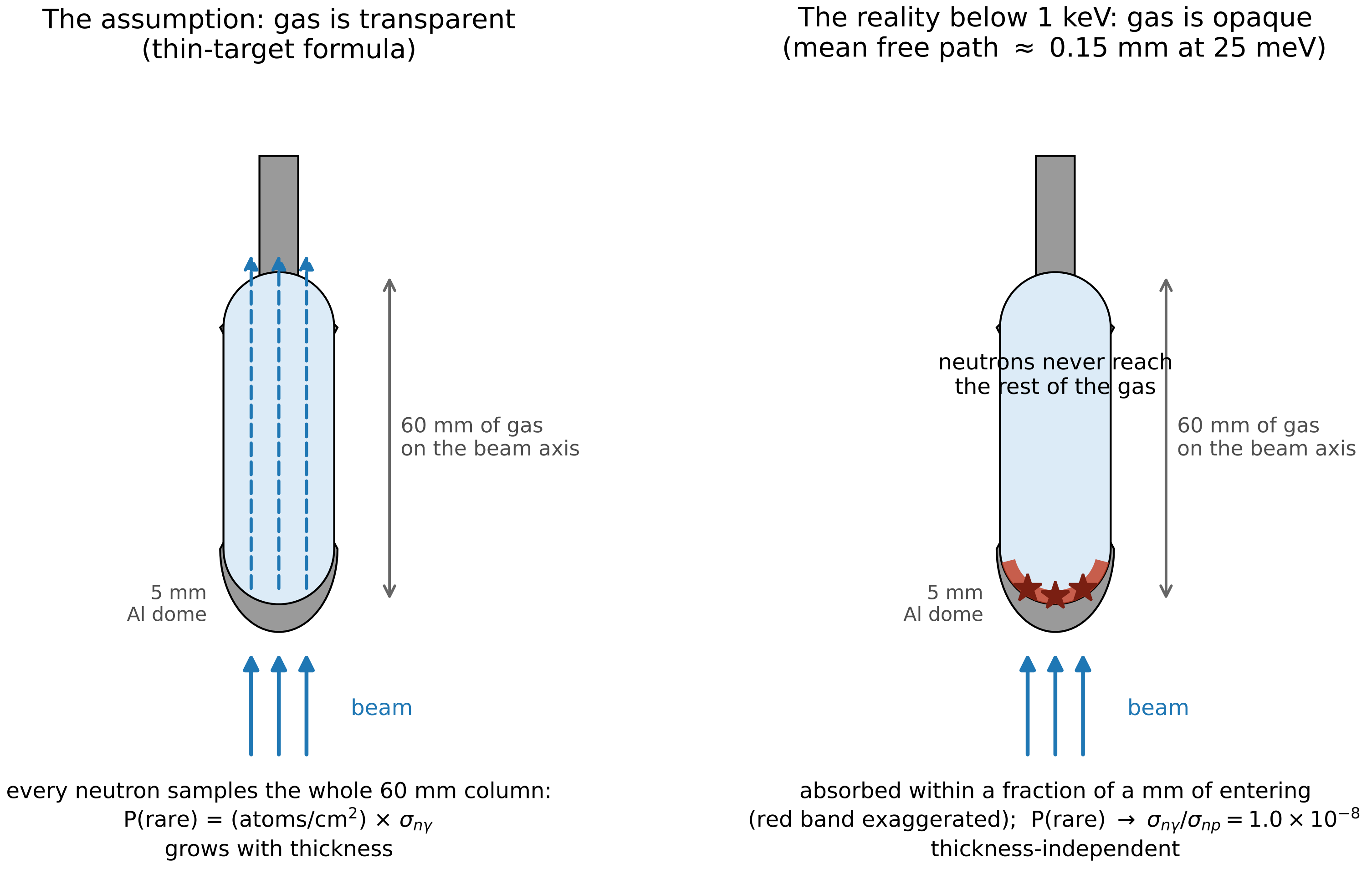

The assumption: gas is transparent
(thin-target formula)
60 mm of gas
on the beam axis
5 mm
Al dome
beam
every neutron samples the whole 60 mm column:
P(rare) = (atoms/cm^2) × $\sigma_{n\gamma}$
grows with thickness
The reality below 1 keV: gas is opaque
(mean free path $\approx$ 0.15 mm at 25 meV)
neutrons never reach
the rest of the gas
60 mm of gas
on the beam axis
5 mm
Al dome
beam
absorbed within a fraction of a mm of entering
(red band exaggerated); P(rare) $\rightarrow$ $\sigma_{n\gamma}/\sigma_{np} = 1.0 \times 10^{-8}$
thickness-independent